# P-adic Numbers

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## 1 Introduction

This is a paper on $p$-adic numbers, a type of number that is of great relevance in number theory, and can also be thought of from an analytical perspective. All the sections up to section 5 focus on the number theoretic aspects of $p$-adics, while section 6 and on discuss them more analytically.

## 2 Bases

$p$-adic arithmetic works very similarly to arithmetic in different number bases, so we will first review bases. Typical numbers are written in base-10. The rightmost digit of a base-10 integer has a place value of 1, the digit directly to the left of that has a place value of 10, and so on. Furthermore, each of these digits are integers from 0 to 9. The value of a base-10 integer is then the sum of each digit multiplied by its place value. For example, $345 = 3 \cdot 100 + 4 \cdot 10 + 5 \cdot 1$. In general, if a number is written without a subscript for a base, it is written in base-10 unless it is clear otherwise. As a side note, base 2 is also called binary.

This can be generalized to other bases. In particular, if $n$ is a positive integer, it can be uniquely expressed in base $b$ as $a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0$ where $0 \leq a_i < b$ for $i = 0, 1, \ldots, k$. While we will not formally prove the uniqueness of this representation, Section 2.1 goes over the general process of finding the base $b$ representation of an integer. For ease of notation, $n$ can be written here in base $b$ as $\overline{a_k a_{k-1} \ldots a_1 a_0}_b$. The overline typically isn't needed when expressing numbers whose digits aren't variables.

Notice how numbers in base $b$ are analogous to numbers in base-10. Each digit is an integer between 0 and $b-1$ inclusive and each digit's place value corresponds to some power of $b$. In base 10, this is the case as well because 10, 100, 1000, etc. are all powers of 10.

### 2.1 Arithmetic

Usual arithmetic for numbers in different bases is analogous to that of base 10. Here are some examples:

$$
\begin{array}{cccc}
 & 1 & 1 & \\
 & 2 & 4 & 5_7 \\
+ & 1 & 6 & 6_7 \\
\hline
 & 4 & 4 & 4_7
\end{array}
\qquad
\begin{array}{cccc}
 & 1 & 14 & \\
 & \cancel{2} & \cancel{4} & 6_7 \\
- & 1 & 6 & 5_7 \\
\hline
 & 0 & 5 & 1_7
\end{array}
\qquad
\begin{array}{cccccc}
 & & & 2 & 4 & 5_7 \\
 & & \times & 1 & 6 & 5_7 \\
\hline
 & & 1 & 6 & 2 & 4_7 \\
 & 1 & 1 & 5 & 2 & 0_7 \\
+ & 2 & 4 & 5 & 0 & 0_7 \\
\hline
 & 4 & 1 & 2 & 4 & 4_7
\end{array}
$$

Notice how all remainders, carries, etc. are happening in the same base that the overall operation is taking place in.

### 2.2 Conversion Between Bases

Finally, we will look at conversions between bases. Let's say we want to convert the number 789 from base 10 to base 2. We will first find the last digit. In any base-2 number, if the last digit is 0, the whole number is even, and otherwise it is odd. This is because all other place values are divisible by 2. Therefore, simply by dividing 789 by 2 and finding the remainder, we get the last digit of 789 in binary, which is 1. However, if we keep the quotient of this divison (which is 394) and divide by 2 again, we can find the second to last digit (which is 0). Here is the full process of converting 789 to binary.

$$\begin{aligned}
789 &= 2 \cdot 394 + 1 \implies \mathbf{1} \\
394 &= 2 \cdot 197 + 0 \implies \mathbf{0} \\
197 &= 2 \cdot 98 + 1 \implies \mathbf{1} \\
98 &= 2 \cdot 49 + 0 \implies \mathbf{0} \\
49 &= 2 \cdot 24 + 1 \implies \mathbf{1} \\
24 &= 2 \cdot 12 + 0 \implies \mathbf{0} \\
12 &= 2 \cdot 6 + 0 \implies \mathbf{0} \\
6 &= 2 \cdot 3 + 0 \implies \mathbf{0} \\
3 &= 2 \cdot 1 + 1 \implies \mathbf{1}
\end{aligned}$$

At this point, we stop dividing because we have reached a quotient of 1, which is less than 2, so we can no longer divide it. We have to read this string of digits from the bottom to the top because we said the first divison would give us the last digit, the second division would give us the second to last digit, and so on. Therefore, we have $789 = 100010101_2$. Using a similar process, we can convert numbers between any bases.

## 3 10-adics

The general idea of 10-adic numbers is taking a base-10 number and letting there be infinite digits going on to the left. Recall that when we use decimals, we let there be infiinte digits going on to the right. If we have infinite digits going on to the left however, the place values of the digits can get infinitely large, and the number is no longer finite. However, we will ignore this problem for now and make some meaning out of these numbers with infinite digits to the left.

Some examples of 10-adic integers are $\ldots 111$, $\ldots 1212$, and $\ldots 34345$. Notice that all these 10-adic integers have a sequence of digits which repeats infinitely to the left. We can write 10-adic integers with an infinitely repeating sequence of digits more concisely by putting a bar over the repeated part, just as we would for repeating decimals. Using this notation, the 3 previously mentioned 10-adic integers would be written as $\ldots 1\overline{1}$, $\ldots 12\overline{12}$, and $\ldots 34\overline{34}5$, or $\ldots\overline{1}$, $\ldots\overline{12}$, and $\ldots\overline{34}5$.

## 3.1 Arithmetic

Just like with regular base-10 integers, we can add 10-adic integers. When adding, we would just have to add digits infinitely to the left. For example, adding the two 10-adic integers $\ldots\overline{1}$ and $\ldots\overline{12}$, we get:

$$\begin{array}{rccccc} & \ldots & 1 & 1 & 1 & 1 \\ + & \ldots & 1 & 2 & 1 & 2 \\ \hline & \ldots & 2 & 3 & 2 & 3 \end{array}$$

We could imagine doing this addition infinitely to the left, and we would always get pairs of the digits 2 and 3, so the result of this addition is the 10-adic integer $\ldots\overline{23}$. We can also perform multiplication as normal:

$$\begin{array}{rcccccccc} & & & \ldots & 1 & 2 & 1 & 2 & 1 & 2 \\ & & \times & \ldots & 1 & 1 & 1 & 1 & 1 & 1 \\ \hline & & & \ldots & 1 & 2 & 1 & 2 & 1 & 2 \\ & & \ldots & 1 & 2 & 1 & 2 & 1 & 2 & 0 \\ & \ldots & 1 & 2 & 1 & 2 & 1 & 2 & 0 & 0 \\ + & & \vdots & \vdots & \vdots & \vdots & \vdots & 0 & 0 & 0 \\ \hline & \ldots & 3 & 0 & 9 & 8 & 6 & 5 & 3 & 2 \end{array}$$

## 3.2 Negatives

Interestingly, in the 10-adics, negative signs are not needed. First of all, when we take a negative number $x$, we are really only finding the number $y$ such that $x + y = 0$. Suppose we were trying to find the negative of 1 in the 10-adics. If the negative of 1 is $\ldots\overline{a_3a_2a_1a_0}$, then we know that $\ldots\overline{a_3a_2a_1a_0} + 1 = 0$. Using this information, we can find $\ldots\overline{a_3a_2a_1a_0}$ digit by digit as follows:

$$\begin{array}{rcccccc} & \ldots & a_4 & a_3 & a_2 & a_1 & a_0 \\ + & \ldots & 0 & 0 & 0 & 0 & 1 \\ \hline & \ldots & 0 & 0 & 0 & 0 & 0 \end{array}$$

For the last digit of the sum to be 0, $a_0$ must be 9. Putting $a_0 = 9$ into our addition gives:

$$\begin{array}{rcccccc} & & & & & 1 & \\ & \ldots & a_4 & a_3 & a_2 & a_1 & 9 \\ + & \ldots & 0 & 0 & 0 & 0 & 1 \\ \hline & \ldots & 0 & 0 & 0 & 0 & 0 \end{array}$$

Once again, we must have $a_1 = 9$ due to the carry. Repeating this process, we get that all the digits of $\ldots\overline{a_3a_2a_1a_0}$ are 9, so $-1 = \ldots 999$ in the 10-adics. We We can apply this procedure to any 10-adic number we want to get its negative. For example, you can try finding the negatives of $\ldots\overline{236}9$ and $\ldots\overline{2}598$. First, we will introduce a method that calculates negatives of 10-adics much easier.

To find the negative of a 10-adic integer $\ldots\overline{a_3a_2a_1a_0}$, first find $9 - a_i$ for each digit $a_i$. Since $a_i$ is a digit and is in the range 0 to 9 inclusive, $9 - a_i$ will also be in that range, so it will also be a digit. Therefore, the number $\ldots\overline{(9-a_3)(9-a_2)(9-a_1)(9-a_0)}$ is a valid 10-adic (this is known as the 9's complement of our original number). When you add $\ldots\overline{a_3a_2a_1a_0}$ and $\ldots\overline{(9-a_3)(9-a_2)(9-a_1)(9-a_0)}$ digit, you get a 9 for every digit, so the sum of those two 10-adics is $\ldots 999 = -1$ as we have seen.

In other words:

$$\dots\overline{a_3a_2a_1a_0} + \dots\overline{(9-a_3)(9-a_2)(9-a_1)(9-a_0)} = -1$$
$$\implies -(\dots\overline{a_3a_2a_1a_0}) = \dots\overline{(9-a_3)(9-a_2)(9-a_1)(9-a_0)} + 1$$

Thus, we have found the negative of $\dots\overline{a_3a_2a_1a_0}$. It is 1 plus the 9's complement of $\dots\overline{a_3a_2a_1a_0}$. Here's how we can use this method to find the negatives of the 2 examples mentioned earlier, $\dots\overline{23}69$ and $\dots\overline{25}98$.

For the first one, the 9's complement is $\dots\overline{76}30$. Adding 1 to this, we get $\dots\overline{76}31$.
For the second one, the 9's complement is $\dots\overline{74}01$. Adding 1 to this, we get $\dots\overline{74}02$.

Now that we can find the negative of any 10-adic integer, we can subtract any two 10-adic integers. For example, if we wanted to do $\dots\overline{23}69 - (\dots\overline{25}98)$, it would be equivalent to $\dots\overline{23}69 + (-\dots\overline{25}98)$. As we calculated earlier, $-\dots\overline{25}98$ is $\dots\overline{74}02$. Thus, our problem becomes $\dots\overline{23}69 + (\dots\overline{74}02)$,which becomes:

$$\begin{array}{ccccccccccccccc}
 & & & 1 & 1 & & & 1 & 1 & 1 & 1 & & & 1 & 1 \\
 & \dots & 6 & 2 & 3 & 6 & 2 & 3 & 6 & 2 & 3 & 6 & 2 & 3 & 6 & 9 \\
+ & \dots & 7 & 4 & 7 & 4 & 7 & 4 & 7 & 4 & 7 & 4 & 7 & 4 & 9 & 8 \\
\hline
 & \dots & 3 & 7 & 1 & 0 & 9 & 8 & 3 & 7 & 1 & 0 & 9 & 8 & 6 & 7
\end{array}$$

Notice that we can now see a repetition in the sum up to 6 digits. Therefore, the desired answer to our original subtraction problem is $\dots\overline{371098}67$

### 3.3 Multiplicative Inverses

Some 10-adic integers also have a multiplicative inverse within the 10-adic integers. For example 1 and -1 ($= \dots 999$) are obviously multiplicative inverses of themselves. To explore this idea of multiplicative inverses, let us find the multiplicative inverse of 3 in the 10-adics.

First of all, multiplicative inverses are numbers such that when multiplied, result in 1. Therefore, we want to find $\dots\overline{a_3a_2a_1a_0}$ such that $3(\dots\overline{a_3a_2a_1a_0}) = 1$. We can now set up the multiplication as follows:

$$\begin{array}{cccccc}
 & \dots & a_3 & a_2 & a_1 & a_0 \\
\times & \dots & 0 & 0 & 0 & 3 \\
\hline
 & \dots & 0 & 0 & 0 & 1
\end{array}$$

For the last digit of the product to be 1, $3a_0$ must have a remainder of 1 mod 10, which implies that $a_0 = 7$. With this value of $a_0$, our multiplication then becomes:

$$\begin{array}{cccccc}
 & & & & 2 & \\
 & \dots & a_3 & a_2 & a_1 & 7 \\
\times & \dots & 0 & 0 & 0 & 3 \\
\hline
 & \dots & 0 & 0 & 0 & 1
\end{array}$$

This means $3a_1$ has a remainder of 8 mod 10, so $a_1 = 6$. When we plug this into the multiplication again, the next carry is going to be another 2, so $a_2$ will also be 6, and every digit besides $a_0$ will be 6. In other words, we get:

$$\begin{array}{cccccc}
 & \dots & 6 & 6 & 6 & 7 \\
\times & \dots & 0 & 0 & 0 & 3 \\
\hline
 & \dots & 0 & 0 & 0 & 1
\end{array}$$

The digits $\ldots 6667$ are very familiar. In fact, they are just the digits of $\frac{2}{3}$ expressed as a decimal, rounded. However, if we negative the 10-adic integer $\ldots 6667$, we get $\ldots 3333$, which are exactly the digits of $\frac{1}{3}$ in its decimal representation. This is not a coincidence, and we will investigate why in the next section.

### 3.4 (Non)existence of Multiplicative Inverses

Unlike in our above example, multiplicative inverses don't always exist in the 10-adics. For example, 10 doesn't have a 10-adic inverse. This is because no matter what you multiply by 10, there will always be a 0 in the ones place and not a 1.

In general, any 10-adic with a last digit of 10 will not have a multiplicative inverse. Furthermore, 10-adics with a last digit of 2 or 5 won't have multiplicative inverses. This is because when you multiply such a number by any other 10-adic number, the last digit of the product will retain either a factor of 2 or 5, and thus the last digit will not be 1. This is directly due to the fact that 2 and 5 are factors of 10.

Instead, what we need for multiplicative inverses is a last digit that has a multiplicative inverse mod 10. Multiplicative inverses are numbers that multiply to something congruent to 1 mod 10, such as the 3 and 7 from the example in the previous section. If in the 10-adic number $\ldots a_3a_2a_1a_0$, $a_0$ has a multiplicative inverse mod 10, then we can have the multiplicative inverse of the entire 10-adic number end in the multiplicative inverse of $a_0$, allowing the product to have a final digit of 1.

As it turns out, a 10-adic number having a last digit that has a multiplicative inverse mod 10 is sufficient for it to have a 10-adic inverse, which we summarize in the following theorem:

**Theorem 3.1.** If in the 10-adic integer $a = \ldots a_3a_2a_1a_0$, $a_0$ has an inverse mod 10, $\ldots a_3a_2a_1a_0$ has an inverse in the 10-adics.

*Proof.* As discussed previously, the last digit of the inverse of $a$ must be the inverse of $a_0$ mod 10, which we will denote as $a_0^{-1}$. Let the rest of the digits be denoted by $b_i$, making the inverse $b = \ldots b_3b_2b_1a_0^{-1}$. We now have to solve for the $b_i$'s. Here is the column multiplication for $ab$ ($b$ is the inverse):

$$\begin{array}{rccccc} & & c_3 & c_2 & c_1 & \\ & \ldots & b_3 & b_2 & b_1 & a_0^{-1} \\ \times & \ldots & a_3 & a_2 & a_1 & a_0 \\ \hline & \ldots & 0 & 0 & 0 & 1 \end{array}$$

Notice that all the carries in this multiplication are denoted by $c_i$. The first carry, $c_1$ is already determined by the values of $a_0$ and $a_0^{-1}$, so we can definitely use it to find $b_1$.

For the second digit of the product to be 0, we must have $a_0b_1 + a_1a_0^{-1} + c_1 \equiv 0 \pmod{10}$. Solving for $b_1$, we get $b_1 \equiv -a_0^{-1}(a_1a_0^{-1} + c_1)$. $a_0^{-1}$ exists by assumption, and we are also assuming we already know $a_1$ and $c_1$. Now that we know $b_1$, we can calculate $c_2$. With a similar calculation as the one for $b_1$, we can find that $b_2 \equiv -a_0^{-1}(a_2a_0^{-1} + a_1b_1 + c_2) \pmod{10}$. We can keep on doing this for each $b_i$, showing that the inverse of $a$ does indeed exist. In fact, we have

$$b_i \equiv -a_0^{-1}\left(a_ia_0^{-1} + c_i + \sum_{k=1}^{i-1} a_kb_{i-k}\right) \mod 10.$$

□

# 4 Duality between 10-adics and Decimals

The main idea of this section is that we can relate 10-adics and decimals to each other and sometimes do algebra with the two by considering both as representing fractions. In other words, both (repeating) 10-adics and decimals both represent specific rational numbers.

## 4.1 Geometric Series

One formula that would be really useful if we could apply to the 10-adics is the infinite geometric series formula, which says:

$$1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

For real numbers, this only holds if $|x| < 1$. Otherwise, the sum doesn't converge. For example, this formula would suggest that $1+2+2^2+\cdots = \frac{1}{1-2} = -1$, but this is ridiculous because the terms in the sum get infinitely large, and can't possible converge to a finite value, much less a negative value. However, we do have the following proposition:

**Proposition 4.1.** If $x$ is a number in any algebraic number system with commutative addition and multiplication, and the sum $1 + x + x^2 + \ldots$ converges to some value in that number system, then it converges to $\frac{1}{1-x}$.

*Proof.* Suppose that $1 + x + x^2 + \ldots$ converges to $S$. Then, we have that

$$\begin{aligned} S = 1 + x + x^2 + \cdots = 1 + x(1 + x + x^2 + \ldots) &= 1 + Sx \\ \implies S(1-x) &= 1 \\ \implies S &= \frac{1}{1-x} \end{aligned}$$

□

"Convergence" and "algebraic number system" here are used quite loosely, but since we are only applying this to the 10-adics (and $p$-adics later), this will suffice.

Now, here is the main claim that will allow us to use infinite geometric series for 10-adics:

**Theorem 4.1.** For a 10-adic integer $x$, the infinite geometric series $1 + x + x^2 + \ldots$ converges if and only if 10 divides $x$.

Here, 10 dividing $x$ is equivalent to $x$ having a 0 as its ones digit. In general, divisibility of a 10-adic by a regular positive integer isn't well-defined unless if it's divisibility by 1, 2, 5, or 10 because those are the factors of 10. For example, it doesn't make sense to ask if a 10-adic integer is divisible by 3.

*Proof.* Rather than providing a rigorous proof of the statement, we will sketch one with some particular examples.

Let's start with the backwards direction. Suppose $x$ is 90, which is divisible by 10. Then, the geometric series $1 + x + x^2 + \ldots$ is $1 + 90 + 90^2 + 90^3 + \cdots = 1 + 90 + 810 + 729000 + \ldots$. Writing this using column addition, we get:

$$\begin{array}{rcccccc} & & & & & 9 & 0 \\ & & & 8 & 1 & 0 & 0 \\ & 7 & 2 & 9 & 0 & 0 & 0 \\ + & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \hline \end{array}$$

The last digit of the final sum is pretty clearly 0 because every term has 0 as its last digit. The second to last digit is pretty clearly 9 because after adding the 9 from 90, the second to last digit of every other number is 0. In fact, for every digit of the final sum, you only have to sum a finite number of digits from the summands in the geometric series. Thus, we can calculate each digit of the final sum if we wanted to, so it must converge.

For the forwards direction, we will do a proof by contrapositive. Suppose that $x$ is a 10-adic integer that isn't divisible by 10, say 2. Then, the geometric series we are interested in is $1 + 2 + 2^2 + \dots$, which is $1 + 2 + 4 + 8 + 16 + 32 + 64 + \dots$. Let's consider only the last digit of this series. To do this, we only need to consider the last digits of each of the terms in the geometric series, so the last digit of $1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + \dots$ (if it exists) is the same as that of $1 + 2 + 4 + 8 + 6 + 2 + 4 + 8 + 6 \dots$.

Notice that these last digits repeat. In this case, the repetition is 2, 4, 8, 6. In general, these last digits will always repeat because of Fermat's Little Theorem. Now, if we take the last digits of the partial sums of the series $1 + 2 + 4 + 8 + 6 + \dots$, it never becomes constant, because the sequence $1, 2, 4, \dots$ never becomes 0. Instead, the last digit only repeats between a couple of values. Therefore, the last digit of the series doesn't converge to one value, so the entire series also can't converge. $\square$

With this, we can calculate things like $1+90+90^2+\dots$. We just get $\frac{1}{1-90} = -\frac{1}{89}$. What this means is that $89(1 + 90 + 90^2 + \dots) = -1$. Conversely, we can also use this to calculate the reciprocal of 89 in the 10-adics. It is just $-(1+90+90^2+\dots)$. Like we said before, all digits of $1+90+90^2+\dots$ can be calculated with enough work, so $\frac{1}{89}$ can be calculated as well.

In the last section, we also saw that $\frac{1}{3} = \dots 6667 = -(\dots 333)$. Notice that

$$\begin{aligned} \dots 333 &= 3(\dots 111) \\ &= 3(1 + 10 + 10^2 + \dots) \\ &= 3(\frac{1}{1-10}) \\ &= 3(-\frac{1}{9}) \\ &= -\frac{1}{3}. \end{aligned}$$

Therefore, $\dots 6667 = -(\dots 333) = -(-\frac{1}{3}) = \frac{1}{3}$, which is the same as what we saw previously.

## 4.2 Calculating Multiplicative Inverses

Using the geometric series formula, we can calculate the reciprocal of any finite 10-adic number. We saw in the previous section how $\frac{1}{89} = -\frac{1}{1-90} = -(1 + 90 + 90^2 + \dots)$. However, we can't apply this to something like $\frac{1}{43}$ because $\frac{1}{43} = -\frac{1}{1-44}$, but by Theorem 4.1, $\frac{1}{1-44}$ cannot be written as a geometric series.

However, we can write $\frac{1}{43}$ as $\frac{1}{43} = \frac{3}{3 \cdot 43} = \frac{3}{129} = -\frac{3}{1-130} = -3(1 + 130 + 130^2 + \dots)$. This is an expression that we can evaluate. In general, we can calculate $\frac{1}{n}$ in the 10-adics if $n$ is relatively prime to 10. This is because we can find an integer $k$ such that $\frac{1}{n} = \frac{k}{nk}$ and $nk \equiv 9 \mod 10$. In fact, we can take $k \equiv 9n^{-1} \mod 10$. Now, since $nk$ is one less than a multiple of 10, we can use our geometric series method to calculate $\frac{1}{nk}$ and thus $\frac{1}{n}$. Here is one example:

$$\frac{7}{49} = -\frac{7}{1-50} = -7(1 + 50 + 50^2 + 50^3 + \dots).$$

We can calculate some of the digits of $1 + 50 + 50^2 + \dots$ as follows:

$$\begin{array}{ccccccccccccc}
 & & & & & & & & & & & & 1 \\
 & & & & & & & & & & & 5 & 0 \\
 & & & & & & & & & 2 & 5 & 0 & 0 \\
 & & & & & & & 1 & 2 & 5 & 0 & 0 & 0 \\
 & & & & & & 6 & 2 & 5 & 0 & 0 & 0 & 0 \\
 & & & & 3 & 1 & 2 & 5 & 0 & 0 & 0 & 0 & 0 \\
 & & 1 & 5 & 6 & 2 & 5 & 0 & 0 & 0 & 0 & 0 & 0 \\
+ & 7 & 8 & 1 & 2 & 5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
 & & & & \dots & 9 & 3 & 8 & 7 & 7 & 5 & 5 & 1
\end{array}$$

Note that we can't find the exact value of any other digits to the left of the 9 just adding the terms listed so far. In particular, when we calculate the digit to the left of 9, there will be another digit from the next power of 50 that we have to consider.

Multiplying our number $\dots 93877551$ by 7, we get the following:

$$\begin{array}{cccccccccc}
 & \dots & 9 & 3 & 8 & 7 & 7 & 5 & 5 & 1 \\
\times & & & & & & & & & 7 \\
\hline
 & \dots & 5 & 7 & 1 & 4 & 2 & 8 & 5 & 7
\end{array}$$

Finally, finding the negative, we get $\dots 42857143$. Interestingly, this number seems very similar to $\frac{1}{7}$ expanded as a decimal, which is 0.142857.

## 4.3 Repeating 10-adics to Fraction

In the last section, we converted a fraction to 10-adic integer (which is in fact repeating, but this will become apparent later). Now, we will do the reverse. Suppose we have the number $x = \dots \overline{897}37$. Then, observe that $1000x = \dots \overline{897}37000$ and that $x$ can be written as $\dots \overline{897}89737$. If we subtract $x$ from this, we get

$$\begin{array}{cccccccc}
 & \dots & \overline{897} & 3 & 7 & 0 & 0 & 0 \\
- & \dots & \overline{897} & 8 & 9 & 7 & 3 & 7 \\
\hline
 & & & 4 & 7 & 2 & 6 & 3
\end{array}$$

Here, we did the subtraction $37000 - 89737 = -52737$, and the found the 10-adic negative of 52737, which is 47263. Therefore, $999x = 1000x - x = 47263 \implies x = \frac{47263}{999}$. Overall, from the last two sections, we can now convert between repeating decimals and rational numbers.

## 4.4 Euler's Doubly Infinite Sum

Calculating fractions in the 10-adics like we did for $\frac{1}{7}$ is quite time-consuming, but there is a faster way. We have seen that repeating 10-adics can be written as fractions, and we know that repeating decimals can also be written as fractions. Therefore, it makes some sense to be able to add 10-adics with decimals; we can just add the corresponding fractions.

Before we start actually adding 10-adics to decimals, let us use the following notation: let $\lambda(x)$ be the fraction corresponding to a repeating 10-adic number $x$ and $\mu(x)$ the fraction corresponding to a repeating decimal. Then, we have the following result:

**Theorem 4.2.** If $x \neq 0$ is an integer divisible by 10, then

$$\mu\left(\sum_{n=1}^{\infty} \frac{1}{x^n}\right) + \lambda\left(\sum_{n=0}^{\infty} x^n\right) = \mu\left(\cdots + \frac{1}{x^2} + \frac{1}{x}\right) + \lambda(1 + x + x^2 + \dots) = 0$$

*Proof.* Since $x$ is an integer divisible by 10, $|x| \geq 10$, so $\left|\frac{1}{x}\right| < 1$, allowing us to use the geometric series formula for the first sum, giving:

$$\mu\left(\sum_{n=1}^{\infty} \frac{1}{x^n}\right) = \mu\left(\frac{1}{1-\frac{1}{x}} - 1\right) = \mu\left(\frac{x}{x-1} - 1\right) = \frac{1}{x-1}.$$

Furthermore, since $x$ is divisible by 10, we can use the geometric series formula from Section 4.1 to obtain

$$\lambda\left(\sum_{n=0}^{\infty} x^n\right) = \lambda\left(\frac{1}{1-x}\right) = \frac{1}{1-x}.$$

Adding the two together, we get

$$\frac{1}{x-1} + \frac{1}{1-x} = \frac{1}{x-1} - \frac{1}{x-1} = 0$$

□

The way that Theorem 4.4 is written is mostly just a way to formalize the following, known as Euler's Doubly Infinite Identity:

$$\sum_{n=-\infty}^{\infty} x^n = \cdots + \frac{1}{x^2} + \frac{1}{x} + 1 + x + x^2 + \cdots = 0$$

One way you can "prove" this is as follows:

$$\begin{aligned}
&S = \cdots + \frac{1}{x^2} + \frac{1}{x} + 1 + x + x^2 + \dots \\
&S = \cdots + x \cdot \frac{1}{x^2} + x \cdot \frac{1}{x} + x \cdot 1 + x \cdot x + x \cdot x^2 + \dots \\
&S = x(\cdots + \frac{1}{x^2} + \frac{1}{x} + 1 + x + x^2 + \dots) = Sx \\
&0 = Sx - S = S(x-1) \implies S = 0
\end{aligned}$$

However, this makes many assumptions about convergence. For example, it would imply that the sum $\cdots + \frac{1}{2^2} + \frac{1}{2} + 1 + 2 + 2^2 + \cdots = 0$, even though it obviously doesn't even converge. Furthermore, one has to define what it means to add a 10-adic and decimal together when writing Theorem 4.4 and what it means for the sum to be 0. The statement of Theorem 4.4, however, avoids this.

On the other hand, Theorem 4.4 motivates us to define numbers with both a 10-adic part and a decimal part. Such a number would take the form $\ldots a_3a_2a_1a_0.b_1b_2b_3\ldots$. In this light, Theorem 4.4 is saying that $\ldots\overline{a_na_{n-1}\ldots a_1a_0}.\overline{a_na_{n-1}\ldots a_1a_0}\cdots = 0$ where the overline indicates repetition and the $a_na_{n-1}\ldots a_1a_0$ is the repeated sequence.

As an example, take $\overline{3434}.\overline{3434}$. This is equal to $34(\ldots\overline{0101}.\overline{0101}\ldots) = 34(\cdots + \frac{1}{100^2} + \frac{1}{100} + 1 + 100 + 100^2) = 0$. Therefore, $\ldots\overline{3434} = -0.\overline{3434}\cdots = -\frac{34}{99}$. In a similar fashion, we can calculate $\frac{1}{7}$ as we teased earlier in the section. In particular, we know $\frac{1}{7} = 0.\overline{142857}$ as a decimal, and that $\overline{142857}.\overline{142857} = 0$, so $\overline{142857} = -0.142857 = -\frac{1}{7}$. Therefore, $\frac{1}{7} = -\overline{142857} = \overline{285714}3$

# 5 Orders, Primitive Roots, and Decimal Expansions

Now that we know that repeating 10-adics, repeating decimals, and fractions are all related to each other, the question arises as to how we can tell how long the repeating part of a decimal or 10-adic representation of a fraction is. To do that, we need to use the notion of order mod $n$.

## 5.1 Orders and Primitive Roots

Mod $n$ for any positive integer $n$, if you continually add any number to itself, you eventually get 0 mod $n$, the additive identity. For example, $6 + 6 + 6 + 6 \equiv 0 \mod 8$. If we instead consider continually multiplying a number itself mod $n$, the situation is slightly different and a little more interesting. To parallel the case where we continually added a number to itself and got the additive identity, we would expect repeated multiplication to eventually give us 1 mod $n$, the multiplicative identity, and usually it does. For example, consider 3 mod 7:

$$\begin{aligned} 3^2 &\equiv 3 \cdot 3 \equiv 2 \mod 7 \\ 3^3 &\equiv 2 \cdot 3 \equiv 6 \mod 7 \\ 3^4 &\equiv 6 \cdot 3 \equiv 4 \mod 7 \\ 3^5 &\equiv 4 \cdot 3 \equiv 5 \mod 7 \\ 3^6 &\equiv 5 \cdot 3 \equiv 1 \mod 7 \end{aligned}$$

Since $3^6$ is the lowest power of 3 congruent to 1 mod 7, we say that the order of 3 mod 7 is 6. The formal definition of an order is as follows:

**Definition 5.1.** If $\gcd(a, n) = 1$, then the order of $a$ mod $n$ is the smallest positive integer $k$ such that $a^k \equiv 1 \mod n$

The reason we must impose the relatively prime condition in the definition is because numbers fail to have an order otherwise. For example, consider 2 mod 10:

$$\begin{aligned} 2^2 &\equiv 2 \cdot 2 \equiv 4 \mod 10 \\ 2^3 &\equiv 4 \cdot 2 \equiv 8 \mod 10 \\ 2^4 &\equiv 8 \cdot 2 \equiv 6 \mod 10 \\ 2^5 &\equiv 6 \cdot 2 \equiv 2 \mod 10 \end{aligned}$$

At this point, we have come back to 2 mod 10, and multiplying by 2 anymore will just keep us in this loop. The problem here is that 2 and 10 aren't relatively prime. Therefore, the factor of 2 in 2 is never lost mod 10, and since 1 does not have a factor of 2, no power of 2 can be 1 mod 10. On the other hand, if two positive integers $a$ and $n$ are relatively prime, then $a$ does have an order mod $n$, which is due to Euler's Theorem, which states:

**Theorem 5.1.** If $\gcd(a,n)=1$, then $a^{\phi(n)} \equiv 1 \mod n$ where $\phi(n)$ is the Euler Totient Function, defined by the number of positive integers less than or equal to $n$ relatively prime to $n$.

Euler's Theorem also leads to the following proposition:

**Proposition 5.1.** If $\gcd(a,n)=1$ and $m$ is the order of $a$ mod $n$, then $m|\phi(n)$.

*Proof.* we have $a^m \equiv a^{\phi(n)} \equiv 1 \mod n$. We can write $\phi(n)$ in terms of a quotient remainder when dividing by $m$. In particular, we can let $\phi(n) = mk + r$ where $0 \leq r < m$. Therefore, $a^{\phi(n)} \equiv a^{mk+r} \equiv (a^m)^k a^r \equiv a^r \equiv 1 \mod n$. If $r > 0$, then $m$ is no longer the smallest positive integer such that $a^m \equiv 1 \mod n$, so $m$ would not be the order of $a$ in that case. Therefore, we must have $r = 0$, and thus $\phi(n) = mk \implies m|\phi(n)$ □

When the order of $a$ mod $n$ and $\phi(n)$ are equal, then we say that $a$ is a primitive root mod $n$. One example of a primitive root is 2 mod 11. Note that $\phi(11) = 11 - 1 = 10$ because 11 is a prime.

$$\begin{aligned}
2^1 &\equiv 2 \mod 11\\
2^2 &\equiv 4 \mod 11\\
2^3 &\equiv 8 \mod 11\\
2^4 &\equiv 5 \mod 11\\
2^5 &\equiv 10 \mod 11\\
2^6 &\equiv 9 \mod 11\\
2^7 &\equiv 7 \mod 11\\
2^8 &\equiv 3 \mod 11\\
2^9 &\equiv 6 \mod 11\\
2^{10} &\equiv 1 \mod 11
\end{aligned}$$

Notice how each number from 1 to 10 is included in the list of powers of 2 mod 11. This should make sense because if two powers of 2, say $2^a$ and $2^b$ were congruent such that $1 \leq a < b \leq 10$ and $2^a \equiv 2^b \mod 11$, then we would have $2^{b-a} \equiv 1 \mod 11$, and since $0 < b - a < 10$, we would have that 10 is not actually the order of 2 mod 11. Therefore, primitive roots, if they exist, can be thought of as generating all the numbers mod $n$ that are relatively prime to $n$ by multiplication.

## 5.2 Repeats in Decimal Expansions

So, why are orders and primitive roots important to repeating 10-adics and decimals? It is because of the following theorem:

**Theorem 5.2.** If $\gcd(n,10)=1$ and $m$ is the order of 10 mod $n$, then the repeating part of the decimal expansion of $\frac{1}{n}$ has $m$ digits.

Notice how the condition of relatively primeness is needed here to ensure that 10 has an order mod $n$.

*Proof.* We will demonstrate a particular example rather than providing a rigorous proof. Consider $\frac{1}{7}$. We can obtain its decimal expansion via the long division below:

$$
\begin{array}{r}
0.\overline{142857}\\
7\,\overline{)1.000000}\\
7\phantom{00000}\\
\hline
30\phantom{0000}\\
28\phantom{0000}\\
\hline
20\phantom{000}\\
14\phantom{000}\\
\hline
60\phantom{00}\\
56\phantom{00}\\
\hline
40\phantom{0}\\
35\phantom{0}\\
\hline
50\\
49\\
\hline
1
\end{array}
$$

What we want to focus on are the results of each subtraction step, which includes 3, 2, 6, 4, 5, and 1 in this case. We obtain the first difference 7, by subtracting 10 by the largest multiple of the 7 less than 10, which is 10%7 (% indicates remainder). Similarly, the next number 2, is the remainder when 3 is multiplied by 10 and then divided by 7, which is $((10\%7) \cdot 10)\%7$. By properties of modular arithmetic, we can write this as $10^2\%7$. Similarly, the next remainder is $10^3\%7$, and so on.

When the remainder becomes 1 again, the decimal starts to repeat because we restart the same division problem as the one we started with. By definition, the number of remainders we go through before we get to 1 is the order of 10 mod 7, and each remainder corresponds to one of the digits in the repeating part of the decimal expansion of $\frac{1}{7}$ $\square$

Since we now know how many digits are in the repeating parts of the decimal representation of $\frac{1}{7}$, we can also find that information for the 10-adic representation. In particular, if $0.\overline{142857} = \frac{1}{7}$, then $\overline{142857}. = -\frac{1}{7}$, and $\frac{1}{7} = \overline{285714}3$. Negating $\overline{142857}.$ did not have any effect on the number of digits on the repeating part of the number; it only added an extra digit not part of the repeating part. Therefore, 5.2 is also true for 10-adics, not just decimals.

## 5.3 Some Example Problems

The following 2 problems demonstrate how useful $n$-adics can be. Here, 2-adics will be used rather than 10-adics, but they utilize the same concepts we've been discussing.

**Problem 1. (AMC 12B 2022 #23)** Let $x_0, x_1, x_2, \ldots$ be a sequence of numbers, where each $x_k$ is either 0 or 1. For each positive integer $n$, define

$$S_n = \sum_{k=0}^{n-1} x_k 2^k$$

Suppose $7S_n \equiv 1 \pmod{2^n}$ for all $n \geqslant 1$. What is the value of the sum

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}?$$

**Solution.** Observe that the $S_n$'s define a 2-adic integer because in binary, the $S_n$'s essentially define infinitely long string of binary digits going off to the left. Let's call this 2-adic number $S$. In other words, $S$ is the (2-adic) limit of $S_n$ as $n$ goes to $\infty$. Furthermore, because $S_n$ has the property that $7S_n \equiv 1 \mod 2^n$, $S$ is the 2-adic inverse of $7 = 111_2$. In other words, $S = \frac{1}{111_2}$. Doing binary long division, we can find that $\frac{1}{111_2} = 0.\overline{001}_2$. This makes sense because the order of 2 mod 7 is 3, so by Theorem 5.2 applied to 2-adics, we should expect to have some factor of 3 as the number of digits in the repeating part. Now, converting to a 2-adic, we get $S = \frac{1}{111_2} = -\overline{001}._2 = \overline{011}1._2$. Now, we essentially have all the $x_k$'s embedded into $S$. In particular, $x_0$ is the last digit of $S$, $x_1$ is the second last, and so on. Finally, if $k > 0$, we have the following:

$$x_k = \begin{cases} 1 & k \equiv 1 \mod 3 \\ 1 & k \equiv 2 \mod 3 \\ 0 & k \equiv 0 \mod 3 \end{cases}$$

This is simply from reading off the repeated digits. Finally, we have $(x_{2019}, x_{2020}, x_{2021}, x_{2022}) = (0, 1, 1, 0)$, giving an answer of $2 + 4 = \boxed{6}$.

**Problem 2. (AIME II 2023 #15)** For each positive integer $n$ let $a_n$ be the least positive integer multiple of 23 such that $a_n \equiv 1 \pmod{2^n}$. Find the number of positive integers $n$ less than or equal to 1000 that satisfy $a_n = a_{n+1}$.

**Solution.** Since each of the $a_i$'s are multiples of 23, we can let $a_i = 23b_i$ for $b_i \in \mathbb{N}$. Since $b_i$ is the inverse of 23 mod $2^i$, like with the previous problem, we can say that the $b_i$'s converge to the 2-adic inverse of 23. Call this limit $b$, so $b = \frac{1}{23} = \frac{1}{10111_2}$ in the 2-adics. We proceed by first doing binary long division to find $\frac{1}{10111_2} = 0.\overline{00001011001}_2$. Therefore, as a 2-adic, $\frac{1}{10111_2} = -\overline{00001011001}._2 = \overline{01111010011}1._2$. Now, each $b_i$ is embedded in this 2-adic number as the $i$ righmost digits. For example, $b_5 = 00111_2$.

If we have the condition that $a_n = a_{n+1}$ as in the problem statement, then $b_n = b_{n+1}$. This can only happen if the $n + 1$st digit of $b$ is 0. In other words, we are looking for the number of zeros in $b$ up to and including the 1001st place (1001 because $n \leq 1000 \implies n + 1 \leq 1001$). Each repeated segment of $b$ has a total of 11 digits, 4 of them being 0. One can check that there are 90 such full-length repeated segments before the 10001st place, and 9 left after that. The 90 repeated segments give $90 \cdot 4$ zeros, and the 9 remaining digits give 3 more zeros (counting through the repeated segment from the right). Thus, we have a total of $90 \cdot 4 + 3 = \boxed{363}$.

## 6 P-adics and Convergence

### 6.1 Why the 10-adics Go Wrong

Consider taking 5, squaring it, and doing so repeatedly. Here are the first few results of this process:

$$5^2 = 25$$
$$25^2 = 625$$
$$625^2 = 390625$$
$$390625^2 = 152587890625$$

Notice how for the most part, each time we square a number, the resulting number has most of the digits of the original number at the end. The only exception is 390625 where its square has 890625. Therefore, we will proceed with only the part that matches with 390625, and keep on proceeding in this way:

$$90625^2 = 8212890625$$
$$8212890625^2 = 67451572418212890625$$
$$67451572418212890625^2 = 4549714621689417981542646884918212890625$$

As we keep on squaring, we get a number $n^2$ which has more and more common digits with $n$. In a weird sense, we are getting a number which is equal to its own square (which is not 0 or 1). In the real numbers, this doesn't make sense. The problem here is that the 10 isn't a prime number, so the 10-adics contain so-called zero divisors. In particular, zero divisors are two numbers, neither of which are 0 but whose product is 0.

If we have a solution to the equation $n^2 = n$ as is sort of happening above, we would have $n^2 - n = n(n-1) = 0$. However, since $n \neq 0, 1$, neither $n$ or $n-1$ is 0, so $n$ and $n-1$ are zero divisors. For those interested, below is a rigorous justification that such an $n$ exists in the 10-adics

We will prove that for every positive integer $k$, there exists a positive integer $n_k$ such that $n_k(n_k - 1) \equiv 0 \mod 10^k$. We can let $n_k = a \cdot 5^k$ such that $a \equiv (5^k)^{-1} \mod 2^k$. Therefore, $n_k - 1 = a \cdot 5^k - 1 \equiv (5^k)^{-1} \cdot 5^k - 1 \equiv 1 - 1 \equiv 0 \mod 2^k$. Therefore, $n_k$ is a number such that its last $k$ digits are the same as the last $k$ digits of its square. Therefore, when taking the limit of $n_k$ as $k$ goes to infinity, we get a 10-adic number which squares to itself. The sequence we were generating above is the same as this sequence $n_k$ except that it skips a few terms.

### 6.2 More Definitions/Theorems

$p$-adics are defined analogously to how the 10-adics are defined, except that the digits are between 0 and $p-1$ inclusive and that $p$ is a prime. We will denote them by $\mathbb{Z}_p$. Addition and multiplication also work in the same column-by-column way as for 10-adics. Now, many things of the things we've proven for 10-adics hold for $p$-adics as well, since 10 is not really a special number. Here is a list of some those theorems but applied to $p$-adics:

**Theorem 6.1.** If in the $p$-adic number $a = \ldots a_3a_2a_1a_0$, $a_0$ has an inverse mod $p$, then $a$ has a n inverse in the $p$-adic integers. In other words, if $a_0 \neq 0$, then $a$ has an inverse in the $p$-adic integers.

**Theorem 6.2.** For a $p$-adic integer $x$, the infinite geometric series $1 + x + x^2 + \dots$ converges to a $p$-adic integer if and only if its last digit is 0.

**Theorem 6.3.** For a $p$-adic integer $x$, let $\lambda(x)$ be the fraction corresponding to the repeating $p$-adic number $x$ and $\mu(x)$ the fraction corresponding to the real number $x$, written in base $p$.

If $x \neq 0$ is an integer divisible by $p$, then

$$\mu\left(\sum_{n=1}^{\infty} \frac{1}{x^n}\right) + \lambda\left(\sum_{n=0}^{\infty} x^n\right) = \mu\left(\dots + \frac{1}{x^2} + \frac{1}{x}\right) + \lambda(1 + x + x^2 + \dots) = 0$$

**Theorem 6.4.** If $\gcd(n, p) = 1$ and $m$ is the order of $p$ mod $n$, then the repeating part of the expansion of $\frac{1}{n}$ in base $p$ and in the $p$-adics has $m$ digits.

## 6.3 P-adic Field

Of particular interest is Theorem 6.2 because it guarantees that there exists an inverse to a $p$-adic integer as long as the last digit is not 0, whereas in the 10-adics, inverses don't exist even if the last digit is 2 or 5. Ideally, we would like to have $p$-adic inverses for all elements, and to do so, we have to introduce the field of $p$-adics.

In particular, the $p$-adic field $\mathbb{Q}_p$ consists of all numbers with a $p$-adic part to the left of the decimal point and a finite part after the decimal point. For example, this would include numbers like $\overline{356}.84576_{11}$ in the 11-adics. The addition of the decimal point effectively allows us to multiply by negative powers of $p$. This also gives us inverses for $p$-adics that had 0's at the end. These 0's represent some power of $p$, say $p^k$ so by just multiplying by $p^{-k}$, we have a number that does have an inverse. For example, $(2^4)^{-1} = 10000_2^{-1} = 0.0001_2 = 2^{-4}$.

Overall, this makes $\mathbb{Q}_p$ a field, which is a set of an algebraic structure with addition and multiplication such that addition and multiplication obey the usual commutative associative, and distributive laws, and such that there exist additive and multiplicative inverses. A more precise definition of a field can be found here: `https://en.wikipedia.org/wiki/Field_(mathematics)`.

Another consequence of $\mathbb{Q}_p$ being a field is that it doesn't have zero divisors. Suppose that $x, y \neq 0$ are $p$-adic numbers. When multiplying them, the last nonzero digit of their product will be the product of the last nonzero digits of $x$ and $y$ mod $p$. Since $p$ is a prime, this cannot be 0.

While all rational numbers are embedded within this field, this field also contains irrational numbers; a number would be irrational in the $p$-adics if it never repeats. For example, $\dots 56295141.3_{11}$ where the digits are the (base 10) digits of $\pi$ reversed, is irrational. This is because every rational number in the $p$-adics does repeat.

## 6.4 P-adic Absolute Value

Just like there is an absolute value for the real numbers, there is a corresponding absolute value for each $p$-adic field. In particular, the $p$-adic absolute value of $x$ is denoted by $|x|_p$ and is defined to be $\frac{1}{p^k}$ where $k$ is the place value of the rightmost nonzero digit of $x$. For example, $|1011000|_2 = \frac{1}{2^3}$ and $|10100.11001|_2 = \frac{1}{2^{-5}} = 2^5$. In other words, if $x = p^k y$ where is $y$ is a $p$-adic whose units digit is nonzero and has nothing to the right of the decimal point, then $|x|_p = \frac{1}{p^y}$. Also the $p$-adic absolute value of 0 is defined to be 0. Using this absolute value, we can define the $p$-adic metric between two $p$-adic numbers $x$ and $y$ as the absolute value of their differences, or $|x-y|_p$. In general terms, a metric is a notion of distance defined a set called a metric space. More details can be found here: https://en.wikipedia.org/wiki/Metric_space. With this, we have the following proposition:

**Proposition 6.1.** The following are true for a $p$-adic absolute value where $x$ and $y$ are $p$-adic numbers:

- $|x-y|_p \geq 0$ with equality if and only if $x = y$
- $|x-y|_p = |y-x|_p$
- $|x+y|_p \leq \max(|x|_p, |y|_p) \leq |x|_p + |y|_p$
- $|xy|_p = |x|_p|y|_p$

*Proof.*

- The first statement is trivial because the $|\cdot|_p$ is always positive by the way it is defined.
- Suppose $x - y = p^k z$ where $z$ has a nonzero units digit and has nothing to the right of the decimal point. Then, $|x-y|_p = \frac{1}{p^k}$. Then, when taking the negative of $z$ to $-z$ will also have a nonzero units digit, so $|y-x|_p = |-(x-y)|_p = |p^k(-z)|_p = \frac{1}{p^k} = |x-y|_p$.
- Suppose $x = p^k z$ and $y = p^j w$ where $z$ and $w$ have nonzero units digits and nothing to the right of the decimal point, so $|x|_p = \frac{1}{p^k}$ and $|y|_p = \frac{1}{p^j}$. Suppose $k \leq j$, so $|x|_p \geq |y|_p$ and $\max(|x|_p, |y|_p) = |x|_p$. When adding $x$ and $y$, we get

  $$x + y = p^k z + p^j w = p^k(z + p^{j-k} w).$$

  Since $j - k \leq 0$, $z + p^{j-k}w$ has nothing to the right of the decimal point. If $z + p^{j-k}w$ has $a$ zeros at the end of it then, $|x+y|_p = \frac{1}{p^{k+a}} \leq \frac{1}{p^k} = \max(|x|_p, |y|_p)$.
- Suppose $x = p^k z$ and $y = p^j w$ where $z$ and $w$ have nonzero units digits and nothing to the right of the decimal point. Then, their product, $xy = p^{k+j} zw$. $zw$ also has a nonzero units digit, so $|xy|_p = \frac{1}{p^{k+j}} = |x|_p|y|_p$.

$\square$

Notice that two numbers are close together by the $p$-adic metric when they have many digits in common, because then their difference wil have many 0's at the end of it, resulting in a small absolute value. For example, $|22232323 - 32332323|_5 = |(222_5 - 323_5) \cdot 5^5|_5 = \frac{1}{5^5}$, which is pretty small.

## 6.5 Convergence

With an absolute value, we can define convergence for sequences of $p$-adic numbers. In particular, if $a_0, a_1, \ldots$ is a sequence of $p$-adic numbers, we say that the sequence converges to the $p$-adic number $a$ and write $\lim_{n\to\infty} a_n$ if $\forall \epsilon > 0$, we can find an $N > 0$ such that for all $n > N$, $|a_N - a| < \epsilon$. In other words, the terms in the sequence gets arbitrarily close to $a$ $p$-adically.

We've already seen one example of $p$-adic convergence with Theorem 4.1. In this case, we saw the convergence of a series rather than a sum. The difference is that the sequence of partial sums was what was converging, where the $N$th partial of sum of the series $\sum_{n=0}^{\infty} a_n$ is $\sum_{n=0}^{N} a_n$. In other words, we had the following:

$$\sum_{n=0}^{\infty} \frac{1}{x^n} = \lim_{N\to\infty} \sum_{n=0}^{N} \frac{1}{x^n} = \frac{1}{1-x}$$

where $x$ was divisible by 10 in the 10-adics (and analogously, by $p$ in the $p$-adics).

This is where $p$-adic series differ very interestingly from real series. For example, consider the harmonic series for the real numbers, which is:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

It can be shown that this doesn't converge, by the following argument:

$$\begin{aligned}
&1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) + \dots \\
\geq &1 + \frac{1}{2} + \frac{2}{4} + \frac{4}{8} + \frac{8}{16} + \dots \\
= &1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots \to \infty
\end{aligned}$$

If we compare this with a series that does converge, such as the geometric series, we notice that for both series, the terms get smaller and smaller; in particular, their absolute values go to 0. In fact, for any convergent series in the real numbers, the absolute values of the sequence must converge to 0, which can be summarized in the following proposition:

**Proposition 6.2.** If $a_0, a_1, \dots, a_n$ is a sequence of real numbers such that $\sum_{n=0}^{\infty} a_n = a$ for some $a \in \mathbb{R}$, then the limit $\lim_{n\to\infty} |a_n| = 0$.

*Proof.* The convergence of the series $\sum a_n$ means that the limit of the partial sums converges. In particular, we have:

$$a = \lim_{N\to\infty} \sum_{n=0}^{N} a_n = \lim_{N\to\infty} \left(a_N + \sum_{n=0}^{N-1} a_n\right) = \lim_{N\to\infty} a_N + \lim_{N\to\infty} \sum_{n=0}^{N} a_n = \lim_{N\to\infty} a_N + a.$$

Reading off the far left and right hand sides of this equation and subtracting $a$ from both sides, we get $\lim_{N\to\infty} a_N = 0$. Then, taking the absolute value on both sides, we get $\lim_{N\to\infty} |a_N| = |0| = 0$, the desired result. □

However, as the harmonic series shows, the converse to Proposition6.5 is not true; not all sequences whose absolute values converge to 0 have a convergent sequence. However, the converse is true for $p$-adics. The justification for this is essentially the same as that of the justification for the backwards direction of Theorem 4.1, but here is the statement of the theorem anyway.

**Theorem 6.5.** Let $a_0, a_1, \dots$ be a sequence of $p$-adic numbers. Then, $\lim_{n\to\infty} a_n = 0$ if and only if $\sum_{n=0}^{\infty} a_n$ converges.

The if and only if part comes from the fact that Proposition 6.5 is also true for $p$-adics.

## 6.6 Exponential Function

In the real numbers, one very important series is the one corresponding to the exponential function $e^x$, also known as $\exp(x)$. We will see if this makes sense for $p$-adic numbers. However, to avoid ambiguity, since $e$ is not necessarily defined in the $p$-adics, we will only refer to the exponential function as $\exp(x)$. It is defined as follows:

$$e^x = \exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

For real numbers $x$, this series always converges, but that is not so for the $p$-adics. For example, consider $\exp(2)$ in the 3-adics. We get the sum

$$\sum_{n=0}^{\infty} \frac{2^n}{n!} = 1 + 2 + \frac{2^2}{2!} + \frac{2^3}{3!} + \dots$$

No term of the sum has a positive power of 3; in fact, many of them contain negative powers of 3. This correlates to the fact that each term doesn't have any 0's in the 3-adics, which means that when considering the partial sums, ones digits (and basically all the other digits) keep changing, and thus this sum can't converge by Theorem 6.5 (the $p$-adic absolute values are always greater than or equal to 1)

On the other hand, if we consider $\exp(3)$ in the 3-adics, it is different. We get the sum:

$$\sum_{n=0}^{\infty} \frac{3^n}{n!} = 1 + 3 + \frac{3^2}{2!} + \frac{3^3}{3!} + \dots$$

Now, all the terms (besides the 1) have a positive power of 3, and in fact each successive term has a higher power of 3. We can count the number of powers of 3's in the term $\frac{3^n}{n!}$ by the difference in the powers of 3 in the numerator and denominator. The number of powers of 3 in the numerator is $n$, and the number of powers of 3 in the denominator is well known to be

$$\left\lfloor \frac{n}{3} \right\rfloor + \left\lfloor \frac{n}{3^2} \right\rfloor + \left\lfloor \frac{n}{3^3} \right\rfloor + \dots \leq \frac{n}{3} + \frac{n}{3^2} + \frac{n}{3^3} + \dots = \frac{n}{2}$$

Overall, the difference is greater than or equal to $n - \frac{n}{2} = \frac{n}{2}$, which is increasing.

Therefore, $\exp(3)$ converges in the 3-adics, and with a similar analysis, $\exp(6), \exp(9), etc$ also exist in the 3-adics. In general, for an odd prime $p$ and an integer $n$, $\exp(n)$ in the $p$-adics exists if and only if $p|n$. The reason that $p = 2$ is an exception is that when finding the number of factors of 2 in $n!$, we get

$$\left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{2^2} \right\rfloor + \left\lfloor \frac{n}{2^3} \right\rfloor + \dots \leq \frac{n}{2} + \frac{n}{2^2} + \frac{n}{2^3} + \dots = n$$

Therefore, the $p$-adic absolute values of terms in $\exp(2)$ (of the form $\frac{2^n}{n!}$) don't actually have a limit of 0, and thus don't converge.

## 6.7 Natural Logarithm

Similar to $e^x$, the natural log, or $\ln(1+x)$, also has a nice series expansion. In particular:

$$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Just like for $\exp(x)$, for this to be defined $p$-adically, we need the terms to have increasing powers of $p$. For this to occur, we need $x$ to be divisible by $p$. With this, we can define exponentiation of two $p$-adics as $a^b = \exp(b \ln a)$, if it exists. Many other functions that can be written as power series can also be defined on the $p$-adics, albeit with restrictions on when convergence occurs.

# 7 Ostrowski's Theorem

We will finish off with a proof of Ostrowski's Theorem, which essentially says that given an arbitrary absolute value function on the rational numbers, it is either the standard absolute value function for real numbers or a $p$-adic absolute value function. Let us first define what is meant by an absolute value, or norm, which we will define more generally in terms of a field:

**Definition 7.1.** $|\cdot|$ is a norm on field $F$ if it satisfies the following properties:

- $|x| \geq 0$ for all $x \in F$, and equality holds only when $x = 0$.
- $|x + y| \leq |x| + |y|$ for all $x, y \in F$.
- $|xy| = |x||y|$ for all $x, y \in F$.

The first two conditions essentially define a metric, and the third condition gives us a multiplicative property. Furthermore, we define two norms to be equivalent if the following is true:

**Definition 7.2.** Two norms $|\cdot|_1$ and $|\cdot|_2$ on a field $F$ are equivalent if there exists a positive real number $a$ such that for all $x \in F$, $|x|_1 = |x|_2^a$.

Also, from here on, we will denote the usual norm for real numbers as $|\cdot|_\infty$, so $|x|_\infty = x$ if $x \geq 0$, and $|x|_\infty = -x$ if $x < 0$. Furthermore, we define the trivial norm $|\cdot|_T$ by $|x|_T = 0$ if $x = 0$ and $|x|_T = 1$ otherwise. It is easy to see that this is indeed a norm. We say a norm is nontrivial if it is not the trivial norm.

Given a rational number $\frac{a}{b}$ where $a, b \in \mathbb{Q}$, we can write both $a$ and $b$ in terms of primes and then reduce the powers of each prime, allowing us to write $\frac{a}{b}$ as $\pm p_1^{r_1} p_2^{r_2} \dots p_n^{r_n}$ where the $p_i$'s are primes and the $r_i$'s are integers (not necessarily positive). Given a norm $|\cdot|$, we can then write $|\frac{a}{b}|$ as $|\pm p_1^{r_1} p_2^{r_2} \dots p_n^{r_n}| = |p_1|^{r_1} |p_2|^{r_2} \dots |p_n|^{r_n}$. In other words, any norm on the rationals is defined only by its values on the primes. This will be helpful in now proving Ostrowski's Theorem:

**Theorem 7.1.** If $|\cdot|$ is a nontrivial norm on the rationals, then it is either equivalent to $|\cdot|_\infty$ or $|\cdot|_p$ for some prime $p$.

*Proof.* There are two major cases. The first case is when for all positive integers $n$, $|n| \leq 1$, in which case we claim that $|\cdot|$ is equivalent to $|\cdot|_p$ for some prime. The other case is when there exists a positive integer $n$ such that $|n| > 1$, in which case we claim that $|\cdot|$ is equivalent to $|\cdot|_\infty$.

- __Case 1:__ **For all positive integers $n$, $|n| \leq 1$**

  Since $|\cdot|$ is nontrivial, there exists a positive integer $n$ such that $|n| < 1$. Then, we can write $n$ out in terms of its prime factors, giving us $|p_1|^{r_1} |p_2|^{r_2} \dots |p_n|^{r_n} < 1$ for distinct primes $p_i$ and positive integers $r_i$. It follows that one of the $p_i$'s has an absolute value less than 1 under this norm, call it $p$. We now claim that $p$ is the only prime with an absolute value strictly less than 1.

Suppose $q \neq p$ is also a prime such that $|q| < 1$. Then, we can find some integer $k$ such that $|p|^k, |q|^k < \frac{1}{2}$ since $|p|, |q| < 1$. By the Euclidean Algorithm/Bezout's Lemma, we can also find integers $a$ and $b$ such that $ap^k + bq^k = gcd(p^k, q^k) = 1$. Therefore, we have the inequality

$$1 = |1| = |ap^k + bq^k| \leq |a||p^k| + |b||q^k| < \frac{1}{2}|a| + \frac{1}{2}|b| \leq \frac{1}{2} + \frac{1}{2} = 1$$

Note that along the way we used the fact $|a|, |b| < 1$ even though they probably are negative. This is ok because in general, $|x| = |-x|$. Anyway, we see that we get the inequality $1 < 1$, which is absurd, so $p$ is the only prime with absolute value less than 1. All that remains is to show that $|\cdot|$ is equivalent to $|\cdot|_p$. If we let $a = -\log_p |p|$, we get that

$$|p|_p^a = \left(\frac{1}{p}\right)^{-\log_p |p|} = p^{\log_p |p|} = |p|.$$

Also, it is obvious that $|q|_p^a = |q| = 1$ for any other prime $q$, so $|\cdot|$ and $|\cdot|_p$ are equivalent (remember, a norm on the rationals is defined by its values on the primes).

- __Case 2:__ **There exists a positive integer $n$ such that $|n| > 1$**

First of all, it follows that $n > 1$. Our goal is to show that for any positive integer $b$ greater than 1, $|b| > 1$ as well. To do this, we can write $n^k$ in base $b$, where $k$ is an arbitrary positive integer for now. First of all, $|n^k| = |n|^k > 1$. Furthermore, we can write $n^k$ in base $b$ with $m$ digits, giving us

$$n^k = c_0 + c_1 b + c_2 b^2 + \cdots + c_{m-1} b^{m-1}$$

where $0 \leq c_i < b$ and $m$ is a positive integer. For each term $c_i b^i$, we can construct the following inequality: First of all, $c_i < b$, so $|c_i| \leq |1| + |1| + \cdots + |1| = c_i < b$, so $|c_i| \leq b-1$. Furthermore, $|b^i| = |b|^i \leq \max\{1, |b|^{m-1}\}$ (this can be shown by considering $|b| > 1$ and $|b| < 1$). Also, we have the inequality $n^k \geq b^{m-1}$, which after solving for $m$, we get $m \leq 1 + k \log_b n$. Putting all this together, we can form the following inequality involving $|n|^k$:

$$\begin{aligned}
|n|^k &= |c_0 + c_1 b + c_2 b^2 + \cdots + c_{m-1} b^{m-1}| \\
&\leq |c_0| + |c_1||b| + |c_2||b|^2 + \cdots + |c_{m-1}||b|^{m-1} \\
&\leq \underbrace{(b-1)\max\{1, |b|^{m-1}\} + (b-1)\max\{1, |b|^{m-1}\} + \cdots + (b-1)\max\{1, |b|^{m-1}\}}_{\text{m times}} \\
&= m(b-1)\max\{1, |b|^{m-1}\} \\
&\leq (1 + k\log_b n)(b-1)\max\{1, |b|^{m-1}\} \\
&\leq (1 + k\log_b n)(b-1)\max\{1, |b|^{k\log_b n}\}.
\end{aligned}$$

Now, taking the $k$th root on the far left and right sides, we get

$$|n| \leq (1 + k\log_b n)^{\frac{1}{k}} (b-1)^{\frac{1}{k}} \max\{1, |b|^{log_b n}\}.$$

We can now take the limit as $k$ goes to infinity. The only slightly challenging limit here is that of $(1+k\log_b n)^{\frac{1}{k}}$. We can write it as

$$\begin{aligned}
&\lim_{k\to\infty}\left((1+k\log_b n)^{\frac{1}{k\log_b n}}\right)^{\log_b n}\\
=&\lim_{x\to\infty}\left((x+1)^{\frac{1}{x}}\right)^{\log_b n}\\
=&1^{\log_b n}\\
=&1.
\end{aligned}$$

The limit $\lim_{x\to\infty}(x+1)^{1/x}$ can be evaluated by taking the natural logarithm and then using L'hopital's Rule.

Overall, we get $|n|\le\max\{1,|b|^{\log_b n}\}$, which implies that $|b|>1$ because otherwise, we would have $|b|^{\log_b n}\le 1$, and $|n|\le 1$. Therefore, we have proven that every integer greater than 1 has an absolute value greater than 1. We have also shown that $|n|\le\max\{1,|b|^{\log_b n}\}=|b|^{\log_b n}$ because $|b|>1$. This implies the following inequalities after rearranging with logarithm rules:

$$\begin{aligned}
|n| &\le |b|^{\log_b n}\\
\log_{|b|}|n| &\le \log_b n\\
\frac{\log_b |n|}{\log_b |b|} &\le \frac{\log_{|n|} n}{\log_{|n|} b}\\
\frac{\log_b |n|}{\log_b |b|} &\le \frac{\log_b |n|}{\log_n |n|}\\
\log_n |n| &\le \log_b |b|
\end{aligned}$$

All we needed to derive this inequality was the fact that $|n|>1$, but we also know that $|b|>1$, so the opposite inequality is also true. In other words, $\log_b|b|\le\log_n|n|$, which means $\log_n|n|=\log_b|b|$ for all integers $n$ and $b$ greater than 1. Finally, we can let $a$ be this constant value of $\log_n|n|$, and then for all primes $p$, $|p|_\infty^a=p^{\log_p|p|}=|p|$. Since this is true for all primes, we have that $|\cdot|$ and $|\cdot|_\infty$ are equivalent.

$\square$